

Dataset Recap

Saved memory full ⓘ

Student	Hours Studied x_1	Attendance x_2	Final Score y
A	2	60	65
B	4	80	75
C	6	90	85

Step 1: Calculate Means and Center the Data

$$\bar{x}_1 = \frac{2 + 4 + 6}{3} = 4, \quad \bar{x}_2 = \frac{60 + 80 + 90}{3} = 76.67, \quad \bar{y} = \frac{65 + 75 + 85}{3} = 75$$

Now subtract means from each data point:

Student	$x'_1 = x_1 - \bar{x}_1$	$x'_2 = x_2 - \bar{x}_2$	$y' = y - \bar{y}$
A	$2 - 4 = -2$	$60 - 76.67 = -16.67$	$65 - 75 = -10$
B	$4 - 4 = 0$	$80 - 76.67 = 3.33$	$75 - 75 = 0$
C	$6 - 4 = 2$	$90 - 76.67 = 13.33$	$85 - 75 = 10$

Step 2: Compute OLS Coefficients (for reference)

$$\beta_1 = \frac{\sum x'_1 y'}{\sum x'^2_1}, \quad \beta_2 = \frac{\sum x'_2 y'}{\sum x'^2_2}$$

Calculate sums:

$$\sum x'_1 y' = (-2)(-10) + (0)(0) + (2)(10) = 20 + 0 + 20 = 40$$

$$\sum x'^2_1 = (-2)^2 + 0^2 + 2^2 = 4 + 0 + 4 = 8$$

$$\Rightarrow \beta_1 = \frac{40}{8} = 5$$

$$\sum x'_2 y' = (-16.67)(-10) + (3.33)(0) + (13.33)(10) = 166.7 + 0 + 133.3 = 300$$

$$\sum x'^2_2 = (-16.67)^2 + (3.33)^2 + (13.33)^2 = 278 + 11.1 + 177.7 = 466.8$$

$$\Rightarrow \beta_2 = \frac{300}{466.8} \approx 0.6426$$

Step 3: Compute Intercept

$$\beta_0 = \bar{y} - \beta_1 \bar{x}_1 - \beta_2 \bar{x}_2 = 75 - 5 \times 4 - 0.6426 \times 76.67 = 75 - 20 - 49.27 = 5.73$$

Step 4: Lasso Regression — Minimize

$$\text{Loss} = \sum (y_i - \beta_0 - \beta_1 x_1 - \beta_2 x_2)^2 + \lambda(|\beta_1| + |\beta_2|)$$

Set $\lambda = 10$.

Step 5: Calculate Loss for OLS Coefficients

Since OLS perfectly fits the data points (because we have 3 points and 3 parameters), **squared error is zero**.

Penalty term:

$$10 \times (|5| + |0.6426|) = 10 \times 5.6426 = 56.426$$

Total loss = 0 + 56.426 = 56.426



Step 6: Calculate Loss if $\beta_2 = 0$ (Lasso shrinks attendance coefficient to zero)

Keep $\beta_1 = 5$, recalculate intercept:

$$\beta_0 = \bar{y} - \beta_1 \bar{x}_1 = 75 - 5 \times 4 = 75 - 20 = 55$$

Calculate predictions and squared error

Student	$\hat{y} = \beta_0 + \beta_1 x_1$	Actual y	Error	Squared Error
A	$55 + 5 \times 2 = 65$	65	0	0
B	$55 + 5 \times 4 = 75$	75	0	0
C	$55 + 5 \times 6 = 85$	85	0	0

Squared error sum = 0

Penalty term:

$$10 \times (|5| + |0|) = 10 \times 5 = 50$$

Total loss = 0 + 50 = 50



Step 7: Conclusion

Model	Total Loss	Notes
OLS coefficients	56.426	Both features used
Lasso with $\beta_2 = 0$	50	Attendance coefficient shrunk

Lasso prefers the simpler model (with attendance dropped) since total loss is smaller.

Final Lasso Regression Model:

$$\hat{y} = 55 + 5 \times x_1$$

Assumed Model:

We are predicting y from:

- x_1 : e.g., Hours Studied
- x_2 : e.g., Attendance

The prediction equation:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Loss Function Forms (with 2 independent variables):

1. Lasso Regression (L1 Regularization):

$$\text{Loss}_{\text{Lasso}} = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2})^2 + \lambda (|\beta_1| + |\beta_2|)$$

2. Ridge Regression (L2 Regularization):

$$\text{Loss}_{\text{Ridge}} = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2})^2 + \lambda (\beta_1^2 + \beta_2^2)$$

3. Elastic Net Regression (L1 + L2 Combined):

$$\text{Loss}_{\text{Elastic}} = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2})^2 + \lambda [\alpha (|\beta_1| + |\beta_2|) + (1 - \alpha)(\beta_1^2 + \beta_2^2)]$$

Where:

- λ = regularization strength
- $\alpha \in [0, 1]$: controls trade-off between L1 and L2

Step-by-step Ridge Regression example using a student score dataset,

Dataset: Student Score Prediction

We're predicting Final Exam Score (y) based on:

- x_1 : Hours Studied
- x_2 : Attendance %

Student	Hours Studied (x_1)	Attendance (x_2)	Final Score (y)
A	2	60	65
B	4	80	75
C	6	90	85

Step 1: Center the Data

First, calculate means:

$$\bar{x}_1 = 4, \quad \bar{x}_2 = 76.67, \quad \bar{y} = 75$$

Now subtract means:

Student	x'_1	x'_2	y'
A	-2	-16.67	-10
B	0	3.33	0
C	2	13.33	10

Step 2: Compute OLS Coefficients

Use:

$$\beta_j = \frac{\sum x'_j y'}{\sum x'^2_j}$$

For β_1 :

$$\sum x'_1 y' = 40, \quad \sum x'^2_1 = 8 \Rightarrow \beta_1 = \frac{40}{8} = 5$$

For β_2 :

$$\sum x'_2 y' = 300, \quad \sum x'^2_2 = 466.8 \Rightarrow \beta_2 = \frac{300}{466.8} \approx 0.6426$$

Step 3: Compute Intercept

$$\beta_0 = \bar{y} - \beta_1 \bar{x}_1 - \beta_2 \bar{x}_2 = 75 - 5(4) - 0.6426(76.67) \approx 5.73$$

OLS Equation:

$$\hat{y}_{OLS} = 5.73 + 5x_1 + 0.6426x_2$$

✓ Step 4: Apply Ridge Regression with $\lambda = 10$

Ridge Cost Function:

$$\text{Loss} = \sum (y_i - \hat{y}_i)^2 + \lambda(\beta_1^2 + \beta_2^2)$$

OLS fits perfectly (SSE = 0), so:

$$\text{Penalty} = 10(5^2 + 0.6426^2) = 10(25 + 0.413) = 10(25.413) = 254.13$$

Total Loss = 254.13

✓ Try Shrinking Coefficients:

Let's try:

- $\beta_1 = 4, \beta_2 = 0.5$

Recalculate intercept:

$$\beta_0 = 75 - 4(4) - 0.5(76.67) = 75 - 16 - 38.34 = 20.66$$

✓ Try Shrinking Coefficients:

Let's try:

- $\beta_1 = 4, \beta_2 = 0.5$

Recalculate intercept:

$$\beta_0 = 75 - 4(4) - 0.5(76.67) = 75 - 16 - 38.34 = 20.66$$

Now compute predictions:

Student	x_1	x_2	$\hat{y}$	y	Error	SE
A	2	60	$20.66 + 4 \times 2 + 0.5 \times 60 = 20.66 + 8 + 30 = 58.66$	65	6.34	40.2
B	4	80	$20.66 + 16 + 40 = 76.66$	75	1.66	2.75
C	6	90	$20.66 + 24 + 45 = 89.66$	85	4.66	21.7

$$\text{SSE} = 40.2 + 2.75 + 21.7 = 64.65$$

$$\text{Penalty} = 10(4^2 + 0.5^2) = 10(16 + 0.25) = 162.5$$

$$\text{Total Loss} = 64.65 + 162.5 = 227.15$$

Comparison

Model	Equation	Loss
OLS	$\hat{y} = 5.73 + 5x_1 + 0.6426x_2$	254.13
Ridge	$\hat{y} = 20.66 + 4x_1 + 0.5x_2$	227.15

✓ Ridge gives a slightly lower total loss by shrinking the coefficients — helpful for avoiding overfitting in larger datasets.

Elastic Net Regression — Quick Recap

Elastic Net combines both penalties:

$$\text{Loss} = \sum (y_i - \hat{y}_i)^2 + \lambda_1 \sum |\beta_j| + \lambda_2 \sum \beta_j^2$$

Or in common form:

$$\text{Loss} = \text{SSE} + \lambda \left[\alpha \sum |\beta_j| + (1 - \alpha) \sum \beta_j^2 \right]$$

Where:

- λ : overall regularization strength
- α : mixing ratio between Lasso ($\alpha = 1$) and Ridge ($\alpha = 0$)

Problem Setup (Same as Before)

Student	Hours Studied (x_1)	Attendance (x_2)	Final Score (y)
A	2	60	65
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OLS Solution:

$$\hat{y}_{OLS} = 5.73 + 5x_1 + 0.6426x_2$$

Let's try applying **Elastic Net** with:

- $\lambda = 10$
- $\alpha = 0.5$ (equal weight to L1 and L2)

Step 1: Use Shrunk Coefficients

Assume:

- $\beta_1 = 4$
- $\beta_2 = 0.5$
- Recompute $\beta_0 = 75 - 4 \times 4 - 0.5 \times 76.67 = 20.66$

✓ Step 2: Compute Predicted Values

Student	x_1	x_2	$\hat{y} = 20.66 + 4x_1 + 0.5x_2$	y	Error	SE
A	2	60	$20.66 + 8 + 30 = 58.66$	65	6.34	40.2
B	4	80	$20.66 + 16 + 40 = 76.66$	75	1.66	2.75
C	6	90	$20.66 + 24 + 45 = 89.66$	85	4.66	21.7

$$SSE = 40.2 + 2.75 + 21.7 = 64.65$$

✓ Step 3: Compute Elastic Net Penalty

$$\lambda = 10, \quad \alpha = 0.5$$

- $L1 = |\beta_1| + |\beta_2| = 4 + 0.5 = 4.5$
- $L2 = \beta_1^2 + \beta_2^2 = 16 + 0.25 = 16.25$

Elastic penalty =

$$10 \cdot [0.5(4.5) + 0.5(16.25)] = 10(2.25 + 8.125) = 10.375$$

✓ Step 4: Final Elastic Net Loss

$$\text{Total Loss} = SSE + \text{Penalty} = 64.65 + 104.75 = \boxed{169.4}$$

Comparison Table

Method	Coefficients	Penalty Used	Total Loss
OLS	$\beta_1 = 5, \beta_2 = 0.6426$	None	0
Ridge ($\lambda=10$)	$\beta_1 = 4, \beta_2 = 0.5$	L2	227.15
Lasso	$\beta_1 = 5, \beta_2 = 0$	L1	50
Elastic Net	$\beta_1 = 4, \beta_2 = 0.5$	L1 + L2	169.4

✓ Final Elastic Net Equation:

$$\hat{y} = 20.66 + 4x_1 + 0.5x_2$$